

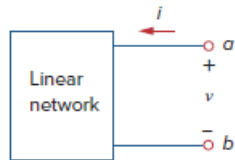
Problem

For the network in Fig. 11.43, assume that the port impedance is

$$\mathbf{Z}_{ab} = \frac{R}{\sqrt{1 + \omega^2 R^2 C^2}} \angle -\tan^{-1} \omega RC$$

Find the average power consumed by the network when $R = 10 \text{ k}\Omega$, $C = 200 \text{ nF}$, and $i = 33 \sin(377t + 22^\circ) \text{ mA}$.

Figure 11.43:



Step-by-step solution

Step 1 of 2

Refer to Figure 11.43 in the textbook.

The current flowing in the circuit is,

$$\begin{aligned} i &= 33 \sin(377t + 22^\circ) \text{ mA} \\ &= 33 \cos(377t - 68^\circ) \text{ mA} \end{aligned}$$

Write the current in phasor form.

$$\mathbf{I} = 33 \angle -68^\circ \text{ mA}$$

The port impedance is,

$$\mathbf{Z}_{ab} = \frac{R}{\sqrt{1 + \omega^2 R^2 C^2}} \angle (-\tan^{-1} \omega RC)$$

Substitute $10 \text{ k}\Omega$ for R , 200 nF for C and 377 rad/s for ω .

$$\begin{aligned} \mathbf{Z}_{ab} &= \frac{10 \text{ k}\Omega}{\sqrt{1 + ((377)^2 (10 \text{ k}\Omega)^2 (200 \text{ nF})^2)}} \angle -\tan^{-1} [(377)(10 \text{ k}\Omega)(200 \text{ nF})] \\ &= \frac{10^4}{\sqrt{1 + (0.754)^2}} \angle -\tan^{-1} (0.754) \\ &= \frac{10^4}{\sqrt{1.5685}} \angle -37^\circ \\ &= 7.9846 \angle -37^\circ \text{ k}\Omega \end{aligned}$$

Step 2 of 2

Calculate the voltage across the port.

$$\begin{aligned} V &= IZ_{ab} \\ &= (33\angle -68^\circ \text{ mA})(7.9846\angle -37^\circ \text{ k}\Omega) \\ &= 263.49\angle -105^\circ \text{ V} \end{aligned}$$

Calculate the average power, P consumed by the network.

$$\begin{aligned} P_{\text{avg}} &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \\ &= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i) \\ &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \\ &= \frac{1}{2} (263.49) (33 \times 10^{-3}) \cos(-105^\circ + 68^\circ) \\ &= (4.347) (0.7986) \\ &= 3.471 \text{ W} \end{aligned}$$

Therefore, the average power, P consumed by the network is 3.471 W.